AN

INTENSHIP REPORT

ON

”Customer Churn Prediction Lifetime Value (LTV) Engine”

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(Computer Engineering)

SUBMITTED BY

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DOMAIN

Data Analytics

ABSTRACT

Customer retention is a critical factor for the success and profitability of any business. Acquiring new customers is often more expensive than retaining existing ones, making it essential for organizations to understand customer behavior and identify customers who are likely to leave. The Customer Churn Prediction Lifetime Value (LTV) Engine is a data analytics and machine learning project developed to predict customer churn and estimate the lifetime value of customers.

This project utilizes customer demographic, service, and billing data to analyze customer behavior patterns and identify key factors influencing churn. Data preprocessing and exploratory data analysis (EDA) are performed using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn. A Random Forest Machine Learning model is implemented to predict whether a customer is likely to churn. Additionally, Customer Lifetime Value (LTV) analysis is conducted to determine the long-term revenue potential of each customer.

The insights generated from the analysis are visualized through interactive Power BI dashboards, enabling businesses to monitor key performance indicators such as total customers, churn rate, customer tenure, monthly charges, and revenue contribution. The project helps organizations make data-driven decisions, improve customer retention strategies, identify high-value customers, and enhance overall business profitability.

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Chapter 1

INTRODUCTION

1.1 Introduction

Customer retention is one of the most important factors for business success. In today's competitive market, losing customers can significantly impact a company's revenue and growth. Therefore, businesses need effective methods to identify customers who are likely to leave and take appropriate actions to retain them.

The Customer Churn Prediction Lifetime Value (LTV) Engine is a data analytics and machine learning project developed to analyze customer behavior, predict churn, and estimate customer lifetime value. The project uses customer demographic, service, and billing information to identify patterns that influence customer retention. Machine learning techniques are applied to predict whether a customer is likely to churn, while LTV analysis helps determine the long-term value of each customer.

The project is implemented using Python for data preprocessing, analysis, and model development, and Power BI for creating interactive dashboards and visualizations. By providing valuable insights into customer behavior and revenue contribution, this system helps organizations improve customer retention strategies, increase profitability, and make data-driven business decisions.

Chapter 2

PROBLEM STATEMENT AND OBJECTIVES

2.1 Title

Customer Churn Prediction Lifetime Value (LTV) Engine

2.2 Problem Statement

Customer churn is a major challenge for businesses, as losing existing customers can lead to reduced revenue and increased customer acquisition costs. Many organizations struggle to identify customers who are likely to leave and understand the factors influencing their decisions. Without proper analysis, it becomes difficult to develop effective retention strategies. Therefore, there is a need for a system that can analyze customer data, predict churn behavior, and estimate customer lifetime value to help businesses improve customer retention and maximize profitability.

2.3 Objectives

-To analyze customer data and identify factors affecting customer churn.
-To predict customers who are likely to leave the company using machine learning techniques.
-To calculate Customer Lifetime Value (LTV) and identify high-value customers.
-To perform data cleaning, preprocessing, and exploratory data analysis.
-To create interactive Power BI dashboards for visualizing customer insights.
-To support data-driven decision-making for customer retention and business growth.
-To help organizations improve profitability by reducing customer churn.

Chapter 3

SCOPE AND RATIONALE OF THE STUDY

3.1 Motivation

-Customer retention is essential for business growth and profitability.
-Acquiring new customers is more expensive than retaining existing customers.
-Businesses often face challenges in identifying customers who are likely to churn.
-Predicting churn helps organizations take preventive actions and improve customer satisfaction.
-Analyzing Customer Lifetime Value (LTV) helps businesses focus on high-value customers and optimize marketing efforts.

3.2 Scope Of Project

The scope of the Customer Churn Prediction Lifetime Value (LTV) Engine includes collecting, processing, and analyzing customer data to identify churn patterns and estimate customer value. The project focuses on customer demographics, service usage, billing information, and contract details to understand factors affecting customer retention.

3.3 Rationale of the Study

In today's competitive business environment, retaining customers is critical for long-term success. Many organizations lose revenue due to customer churn without fully understanding the reasons behind it. By analyzing customer data, businesses can identify customers at risk of leaving and implement effective retention strategies.

Chapter 4

METHODOLOGICAL DETAILS

4.1 Methodological Details

1. Data Collection:
Collected customer data containing demographic, service, and billing information.

2. Data Cleaning:
Removed missing values, duplicates, and inconsistencies to improve data quality.

3. Exploratory Data Analysis (EDA):
Analyzed customer behavior, churn patterns, and key trends using visualizations.

4. Model Development:
Applied the Random Forest Classifier to predict customer churn.

5. LTV Analysis:
Calculated Customer Lifetime Value to identify high-value customers.

6. Dashboard Creation:
Developed an interactive Power BI dashboard to visualize KPIs and business insights.

7. Insight Generation:
Generated actionable insights to improve customer retention and business performance.

Chapter 5

RESULTS

5.1 Results

Figure 5.1: First Slide of Dashboard

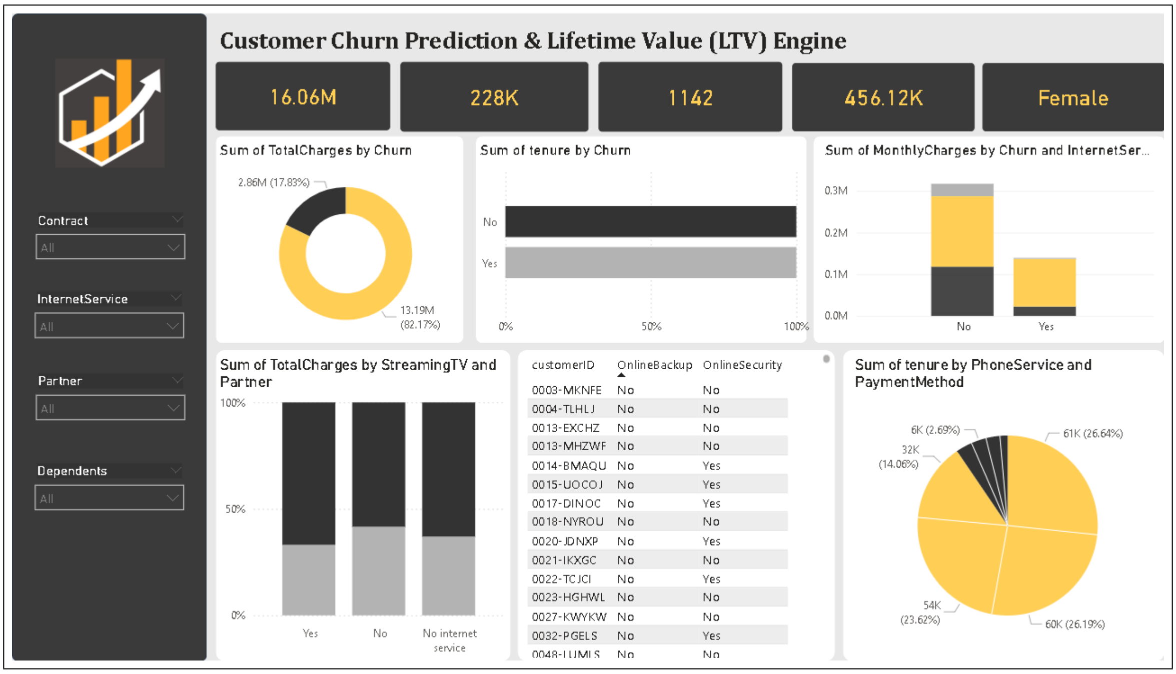

Figure 5.2: Second Slide of Dashboard

5.2 Conclusion

The Customer Churn Prediction Lifetime Value (LTV) Engine successfully analyzes customer data to predict churn and estimate customer value. By using machine learning techniques and Power BI visualizations, the project helps businesses identify customers at risk of leaving and recognize high-value customers. The insights generated support better customer retention strategies, improve decision-making, and enhance overall business profitability. This project demonstrates the effective use of data analytics, machine learning, and business intelligence tools to solve real-world business challenges.

Chapter 6

LIST OF REFERENCES

1. Kaggle: Dataset For Phone Notification Analysis Project.
Website: https://www.kaggle.com

2. ExcelJet: Excel Functions and Data Analysis Tools.
Website:https://exceljet.net/

3. W3School: Python Tutorial – Python Libraries (Pandas, NumPy).
Website: https://www.w3schools.com/python/

4. Microsoft. (n.d.): Power BI Documentation.
website: https://learn.microsoft.com/power-bi/